

Declaration of AI Tools Usage in Scientific Work

such as seminars, research, theses at the Research Group Software Engineering

Title and type of the scientific work (seminar paper, research internship, bachelor's or master's thesis):

Jakarta EE – Examination Project

Submission date:

30.09.2026

1. Overview of the generative AI tools used for this work

Please list all AI tools you used e.g. ChatGPT, DEEPL, MS/GitHub CoPilot, Neuroflash, Grammarly, Google Translate, Perplexity, Elicit, Explainpaper and/or ResearchRabbit :

AI-Tools
1. ChatGPT Plus 2. ChatGPT

2. Responsible use of AI tools

Please confirm the following statements.

Statements on the responsible use of tools	Confirmation: Yes
I/we are informed about the capabilities and limitations of the generative AI tools I/we used.	☒
I/we verified that the results given by the tools are accurate or that I/we corrected them.	☒
I/we acknowledge that the responsibility for the work, e.g. paper/thesis, lies with the author(s), not the tools or anybody else.	☒

3. Detailed activities for which the tools were used

Please indicate which tool you used and to what extent for which activity/step.

Activity/step	Description	AI Tools	Usage Description e.g. type of use, affected sections
Idea generation and conception	- Generation of ideas, research, and questions objectives - Identification and definition of relevant concepts	1) ChatGPT	1) Some initial frontend designs to design our own 2) Ideas for implementation approaches
Literature search and analysis	- Search for relevant literature - Review of potentially relevant literature - Summary of relevant literature	/	/
Methodology	- Search for a appropriate methodology - Design and adaptation of the methodology to your own research (re-search questions)	/	/
Programming	- Creation and documentation of code, algorithms, software, test cases - Testing and debugging existing code, algorithms, software	1) ChatGPT	1) Implementation & debugging of frontend with human set designs and structures 2) Support with individual code optimizations and implementation strategies
Data collection and analysis	- Collection of primary and/or secondary data secondary data - Qualitative data analysis (including Summary and coding/programming). - Quantitative data analysis (including statistics, statistical evaluation) - Mathematical, computational or other formal techniques for modeling, simulation and analysis	/	/
Interpretation and Validation	- Interpretation of the results - Derivation of implications for research and practice - Verification of the general replication/reproducibility of the results and other research results - Structure of the scientific work - Structure of the sections	1) ChatGPT	1) Used to understand errors and verify that the implemented functionality matched the project requirements.
Structuring and Planning of the text/the text sections	(e.g. lists of points per section, bullet points)	/	/
Generation of text/text sections	- Generate text to different topics in different sections of the scientific work (including title and abstract)	1) ChatGPT	1) Partially frontend texts 2) Linguistic refinement of few sections of Wiki
Translation of text/text sections	- Translation of texts written by the authors - Translation of texts written by other authors	1) ChatGPT	1) Translation of i18n into French and Polish

Activity/step	Description	AI Tools	Usage Description e.g. type of use, affected sections
Review and processing of the text/the text sections	- Critical review, feedback or revision of content - Organization or grammar correction of the scientific work - Proofreading. - Reformulation/paraphrasing the text. - Shortening/expanding the text	1) ChatGPT	1) Checking for missed i18n sections 2) Used to clarify Project requirements.
Presentation	- Structuring of the presentation - Presentation with content from the scientific	/	/
Administration of citations/references	- Creation of a reference list - Formatting a reference list	/	/
Other activities	- Others not mentioned above	/	/

4. Is there anything else you can explain and document that your independent, definable performance in creating this work/thesis would highlight or limit?

AI tools were used as supporting tools during the preparation of this work. However, the suggestions and results generated by AI were not adopted without review, but were always critically evaluated, checked for correctness, and adjusted where necessary. The essential considerations, decisions, and the substantive development of the work were carried out independently. AI was therefore used as a supporting tool and not as a substitute for authors' own technical work.

5. Signature(s) of the author(s)

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References:: Gimpel, H., Dilger, P., Lämmermann, L., Urbach, N. (2023). COGNISANCE: Declaration of Generative AI Tool Usage in Higher Education Tests. Research Center for Information Management, May 22, 2023.